21 September 1966

SUPERSEDING
MIL-A-58066(GL)
22 April 1963

MILITARY SPECIFICATION

ALTIMETER, PRESSURE, PARACHUTIST'S

1. SCOPE

1.1 This specification covers one type of single pointer, single dial, pressure altimeter.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Film (Polyethylene Thin Gage)
QQ-P-416	-	Plating, Cadmium (Electrodeposited)
PPP-B-636	-	Box, Fiberboard
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-C-843	-	Cushioning Material, Cellulosic

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-F-495	-	Finish, Chemical, Black, for Copper Alloys
MIL-C-3955	-	Cans, Fiber, Spirally Wound
MIL-E-5272	-	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-25142	-	Luminescent Material, Fluorescent

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

FSC 6610

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MS 15795	-	Washers, Flat, Metal-Round, General Purpose
MS 24123	-	Plate, Identification
MS 33558	-	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS 33586	-	Metals, Definition of Dissimilar
MS 35273	-	Screw, Machine, Drilled Fillister Head, Slotted, Brass, Black Oxide Finish, NC-2A and UNC-2A
MS 35337	-	Washer, Lock, Split, Helical, Light Series
MS 35649	-	Nut, Plain, Hexagon, Machine Screw, NC-2B

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified (see 6.2), prior to starting production, the supplier shall submit to or make ready for the contracting officer or his duly authorized representative 3 or more samples of the finished item for inspection as specified in paragraph 4.2.1 of this specification. The preproduction samples shall be manufactured in the same facilities and by the production tools and methods to be used for manufacture of the production items. The approval of the preproduction samples authorizes the supplier to commence production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification.

3.2 Materials.- Materials used shall conform to the applicable referenced specifications and standards and additional requirements as specified herein. Materials not specifically described or specified herein shall be of the lightest practicable weight and entirely suitable for the purpose intended.

3.2.1 Metals.- Metals used shall be of a corrosion resisting type unless suitably protected against corrosion as specified in 3.5.

3.2.1.1 Dissimilar metals.- Dissimilar metals, as defined in MS33586, shall not be used in intimate contact with each other unless suitably treated to prevent corrosion from galvanic action.

3.2.1.2 Nonmagnetic metals.- Nonmagnetic metals shall be used for all parts of the altimeter except where the use of magnetic materials is specifically indicated due to design requirements.

3.2.2 Cover-glass.- The cover-glass shall be a colorless, transparent polycarbonate plastic material. Minimum thickness shall be 1/8-inch (see 6.3).

3.2.3 Mounting hardware.- The supplier shall furnish the following items of hardware for mounting of the altimeter;

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2 ea Screw, machine, fillister head, size 8-32 conforming to MS 35273, of sufficient length to mount the altimeter to a panel 0.064 inch thick.

2 ea Washer, flat, metal, round, general purpose, conforming to MS 15795-907.

2 ea Washer, lock, split, helical, light series conforming to MS 35337-42.

2 ea Nut, plain, hexagon, machine screw, NC-2B, conforming to MS 35649-85 and finished in accordance with MIL-F-495.

3.2.4 Envelope.- An envelope containing the mounting screws, washers, and nuts shall be furnished by the supplier and packaged with each altimeter. The envelope shall be marked with the following information:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS, WASHERS AND
NUTS

3.3 Design and construction.- The altimeter shall be designed to include the following features: aneroid movement, adjustable zero altitude setting mechanism, a calibration mark on the cover glass (see figure 1) to establish top center position of zero indicator marking on dial at standard mean sea level pressure, and a pointer. The altimeter shall be constructed to with stand the strains, jars, vibrations and conditions incident to shipping, storage, installation and service without failure. The design shall permit overhaul and repair with tools and fixtures normally available commercially and at instrument repair facilities.

3.3.1 Case.- The altimeter case shall conform to the outline dimensions shown on Figure 1, and shall be fabricated of lightweight non-magnetic material. The surface finish shall be smooth, uniform, durable and lusterless black in color.

3.3.2 Mechanism adjustment.- The mechanism shall be provided with means for adjusting or correcting the indications to conform to the performance required herein. The adjustment means shall be designed for setting with commerically available tools.

3.3.3 Zero setting system.- A zero setting system shall be provided to compensate for changes in barometric pressure in order that the altimeter can be set to indicate actual altitude at existing ground level elevation above mean sea level. The zero setting system shall be controlled by rotating the dial with a center mounted setting knob as shown on figure 1. The adjustment requirements shall be as specified in 3.4.5.

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3.3.4 Pointer.- One pointer shall be provided as shown on figure 1. The pointer shall be firmly attached to the mechanism and shall be free to rotate to either side of the zero position with a minimum rotation of thirty degrees in a counter-clockwise direction and with a minimum of one and one-half revolutions required in a clockwise direction before engaging a mechanical stop. Engagement of the pointer mechanism against a mechanical stop shall not cause any change in the calibration of the altimeter. The length of the pointer shall be the maximum length consistent with the size of the altimeter. The pointer shall be coated with fluorescent luminous material, type III, as specified in MIL-L-25142 from the tip to a point 1/16-inch from the center post. The remaining portion of the pointer shall have a lusterless black finish.

3.3.5 Dial.- The dial shall be of metal and have a lusterless black finish. The marking shall be as specified in 3.3.5.1.

3.3.5.1 Marking.- The dial marking shall be as shown on figure 1. The graduation lines shall conform to table I for size. The numerals shall conform to table I for size and MS33558 for form except the numeral 4 shall be open at the top as shown in figure 1. The numerical digits, 500, 1000, and 2000 foot graduation lines, shall be coated with fluorescent luminous material, type III, as specified in MIL-L-25142. The graduations shall be in 500 foot increments with numerals placed at each 2000 foot interval, as shown on figure 1. The numerals shall distinctly indicate the graduations to which they apply.

TABLE I.- Dial marking (Dimensions are in inches)

Markings	Height of Line (Tol ± .010	Width of Line (Tol ± .005)
ALTITUDE SCALE		
Numeral digits: 0, 1, 2, 4, 6, 8	.250	.031
Graduations corresponding to:		
500 ft. increments	.156	.015
1000 ft. increments	.218	.019
2000 ft. increments	.218	.039

3.3.6 Weight.- The weight of the completely assembled altimeter, without mounting screws, washers, and nuts, shall not exceed 0.85 pounds.

3.4 Performance.- The functioning of the altimeter shall conform to the requirements specified below, when tested in accordance with the method of corresponding title set forth in 4.3.1.

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3.4.1 Humidity effects.- Any scale error due to humidity effects shall not exceed the room temperature allowable tolerances at the altitudes specified in table II. The additional criteria for failure specified in MIL-E-5272 shall also apply.

3.4.2 Salt spray effects.- Any scale error due to salt spray effects shall not exceed the room temperature allowable tolerances at the altitudes specified in table II. The additional criteria for failure specified in MIL-E-5272 shall also apply.

3.4.3 Proofload (Mounting lugs).- The mounting lugs shall show no evidence of fracture when subjected to a 175-pound proofload when tested as specified in 4.3.1.4.

3.4.4 Endurance, stability, and shelf life.- Any scale error due to the endurance, stability, and shelf life testing specified in 4.3.1.5 shall not exceed the recorded original room temperature scale error readings (see 3.4.7) at the altitudes specified in table II by more than the values specified in table V.

3.4.5 Zero setting system adjustment.- The torque required to rotate the center mounted setting knob, for adjustment of the altimeter to indicate actual altitude at existing ground pressure, shall be no less than 12 and no more than 24 inch-ounces when tested as specified in 4.3.1.6.

3.4.6 Static balance error.- Any change in pointer indication due to change in altimeter position shall not exceed 100 feet, when tested as specified in 4.3.1.7.

3.4.7 Scale error at room temperature.- Any scale error at room temperature shall not exceed the allowable tolerances at the altitudes specified in table II when tested as specified in 4.3.1.8.

TABLE II.- Scale error tolerances

Standard altitude	Tolerance (+ Feet) at		
	Room Temperature +25°C.	-55°C.	±71°C.
2000	250	500	500
5000	250	500	500
10,000	250	500	500
20,000	400	800	800

3.4.8 Pointer movement.- The movement of the pointer shall be smooth and free from irregular motion during all tests where the pressure is varied at a uniform rate.

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3.4.9 Friction error.- Any change in pointer indication due to friction error shall not exceed the allowable tolerances at the altitudes specified in table III when tested as specified in 4.3.1.10.

TABLE III.- Friction error tolerances

ALTITUDE (feet)	TOLERANCE (± feet)
1,000	100
2,000	100
6,000	150
10,000	200
16,000	200
20,000	300

3.4.10 Hysteresis error.- Any scale error due to hysteresis effects shall not differ from the scale error at room temperature, at the same test points, by more than 200 feet when tested as specified in 4.3.1.11.

3.4.11 Drift error.- Any scale error due to drift shall not exceed a tolerance of plus or minus 100 feet when tested as specified in 4.3.1.12.

3.4.12 Scale error at -55°C.- Any scale error at -55°C. shall not exceed the allowable tolerances at the altitudes specified in table II when tested as specified in 4.3.1.13.

3.4.13 Scale error at +71°C.- Any scale error at +71°C. shall not exceed the allowable tolerances at the altitudes specified in table II when tested as specified in 4.3.1.14.

3.4.14 Damping.- The time from vacuum release in the damping test, until the pointer passes the 3,000 foot indication shall not exceed 2 seconds. Total damping time shall not exceed 3 seconds when tested as specified in 4.3.1.15.

3.5 Protective finish.-

3.5.1 Aluminum alloy parts.- Except for small holes, pipe threads and case inserts, aluminum alloy parts shall be anodized in accordance with the requirements for Type I or II coating, as applicable, of MIL-A-8625.

3.5.2 Steel parts.- Carbon and alloy steel parts, when used, shall be cadmium plated in accordance with the requirements of QQ-P-416 for Type II or III plating, as applicable. The plating class shall be consistent with the design requirements.

3.6 Identification marking.- Each altimeter shall have an identification plate attached thereto. The plate display shall conform to MS24123-8. The

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size, material, location and method of attachment shall be consistent with good practice.

3.7 Workmanship.- The workmanship shall conform to the quality established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies conform to prescribed requirements.

4.2 Inspections.-

4.2.1 Preproduction sample inspection.- Examination and testing of preproduction samples shall be made of completely fabricated altimeters (see 3.1) for all provisions of this specification applicable to material, design, construction, and end product examinations and tests.

4.2.2 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2.1 In-process inspection.- Except as covered by quality assurance provisions of referenced subsidiary specifications and standards, subassemblies shall be inspected and tested at appropriate stages of fabrication, to determine conformance with the applicable requirements of this specification. The Government reserves the right to exclude from consideration for acceptance any components for which in-process inspection has indicated nonconformance.

4.2.3 Inspection of end item.-

4.2.3.1 Sampling for inspection.- Except as otherwise specified herein, sampling and associated procedures for formation of lots for end item and preparation for delivery inspection shall be in accordance with MIL-STD-105. Lots shall be expressed in units of altimeters, or containers, as applicable. The sample unit shall be as specified herein for the applicable examination, inspection or test.

4.2.3.1.1 Inspection levels and acceptable quality levels (AQLs).- Inspection levels and AQLs, expressed in defects per 100 units, shall be as follows:

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	Inspection level	AQLs Major	AQLs Total
For examination in 4.2.3.2	II	.65	2.5 major and minor
For examination in 4.2.3.3	S-2		6.5 one class
For examination in 4.2.3.4	S-2		6.5 one class
For examination in 4.2.3.5	S-2		4.0 one class

4.2.3.2 Examination for visual characteristics.- The sample unit for this examination shall be one altimeter. Characteristics to be examined and the defects applicable thereto shall be as follows:

Examine	Defect	Classification Major	Classification Minor
Assembly (except as otherwise specified herein)	Case color not as specified		X
	Any material or component not finished as specified	X	
	Any component missing, damaged, or malformed	X	
	Any evidence of rust or corrosion		X
	Any rough spots, burrs, or sharp edges	X	
Cover-glass	Not transparent, damaged, any evidence of imperfection	X	
	Calibration mark omitted or not located properly	X	
	Dirt, grease, smudge or other foreign matter inside of cover-glass	X	
Dial and pointer	Any scratches or tool marks through coating to metal		X
	Luminous coating missing, peeling, cracked, not adherent	X	
	Dial graduation markings incomplete, not sharply defined (when examined under ultra-violet light)	X	
	Pointer missing, malformed or damaged	X	
	Pointer loose	X	
Knob	Missing or not securely attached	X	
Identification and special marking	Missing, not legible, incomplete, incorrect	X	

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4.2.3.3 Weight inspection.- An inspection shall be made to determine conformance with the weight requirement specified in 3.3.6. The sample unit shall be one altimeter. Any result indicating weight in excess of that specified shall constitute a defect.

4.2.3.4 Dimensional examination.- Inspection shall be made of the end item for compliance with dimensions and tolerances shown on Figure 1. The sample unit shall be one altimeter. Any evidence of nonconformance shall constitute a defect.

4.2.3.5 Examination of preparation for delivery.- An examination shall be made to determine compliance with the packaging, packing, preservation and marking requirements specified in section 5 of this specification. Defects shall be scored as specified below. Examination shall be made in two phases; first an interior examination, and second an examination of containers fully prepared for delivery. The sample unit for each of the two phases shall be one container fully prepared for delivery with the exception that it need not be sealed or closed.

Examine	Defect
Marking (interior and exterior container)	Omitted; incorrect, illegible; improper size, location, or method of application.
Materials	Component missing, damaged, defective, or not as specified.
Preservation	Incomplete, incorrect or missing.
Workmanship	Improperly positioned in container; improperly cushioned; improper closure of container; loose strapping, bulging or distortion of containers.
Contents (interior and exterior container)	Number per container not as specified. Envelope with hardware missing; quantity of hardware not as specified.

4.2.3.6 Testing of the end item.- The altimeter shall be performance tested in accordance with table IV for the characteristics specified therein. The sample unit for the performance tests shall be one altimeter. The sampling plans shall be as specified in table IV.

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TABLE IV.- Performance testing

Characteristic	Requirement paragraph	Test method paragraph	Record results as	Sampling Plan: Preproduction Sample size	Preproduction Acceptance No.	End item Inspection level	End item AQL	
Humidity effects	3.4.1	4.3.1.2	Scale error and any unacceptable deterioration	3	0	N/A		
Salt spray effects	3.4.2	4.3.1.3	Scale error and any unacceptable deterioration	3	0	N/A		
Proofload (mounting lugs)	3.4.3	4.3.1.4	Pass or fail	3	0	N/A		
Endurance stability, and shelf life	3.4.4	4.3.1.5	Pass or fail	3	0	N/A		
Zero setting system adjustment	3.4.5	4.3.1.6	Torque required to rotate knob in inch-ounces	All	0	100%	Nonconforming altimeters shall be rejected.	
Static balance error	3.4.6	4.3.1.7	Amount of pointer drift in feet	All	0	100%		
Scale error at room temperature	3.4.7	4.3.1.8	Scale error	All	0	100%		
Pointer movement	3.4.8	4.3.1.9	Pass or fail	All	0	100%		
Friction error	3.4.9	4.3.1.10	Pass or fail	All	0	100%		
Hysteresis error	3.4.10	4.3.1.11	Pass or fail	All	0	100%		
Drif__ror	3.4.11	4.3.1.12	Pass or __l	All	0	100%		

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TABLE IV.- Performance testing (cont'd)

Characteristic	Requirement paragraph	Test method paragraph	Record results as	Sampling Plan			
				Preproduction		End item	
				Sample size	Acceptance No.	Inspection level	AQL
Scale error at -55°C.	3.4.12	4.3.1.13	Pass or fail	All	0	II	0.4
Scale error at ±71°C.	3.4.13	4.3.1.14	Pass or fail	All	0	II	0.4
Damping	3.4.14	4.3.1.15	Pass or fail	All	0	II	0.4

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4.3 Tests.-

4.3.1 Performance.-

4.3.1.1 Conditions.- Except as otherwise specified in 4.3.1.4 and 4.3.1.6, the altimeter shall be tested in normal operating position. The tests shall be made at atmospheric pressure (29.92 inches of mercury) and at room temperature (+25°C.) whenever specific pressure and temperature are not specified. When tests are made with atmospheric pressure or room temperature differing from the above values, proper allowance shall be made for the difference from the specified condition. The altimeter shall be lightly tapped or vibrated before test readings are taken except as otherwise specified in 4.3.1.10.

4.3.1.2 Humidity effects.- The altimeter shall be subjected to six cycles of Procedure I type humidity conditioning in accordance with MIL-E-5272, and the test as specified in 4.3.1.8.

4.3.1.3 Salt spray effects.- The altimeter shall be subjected to 50 hours of Procedure I type salt spray conditioning in accordance with MIL-E-5272, and the test as specified in 4.3.1.8.

4.3.1.4 Proofload (mounting lugs).- The altimeter case specified in 3.3.1 with mechanism removed, shall be mounted face downward on the moveable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to the pull strap. A static load of 175 pounds shall be applied along the axis toward the front of the case for a period of one minute. The lugs shall then be examined for any evidence of fracture, deformation, or other failure.

4.3.1.5 Endurance, stability, and shelf life.- The altimeter shall be subjected to and shall meet the requirements of all the individual performance tests. The room temperature scale error readings (see 4.3.1.8) will be referred to later in this test as "the original scale error readings." The instrument shall be subjected to a pressure cycling process consisting of 2,000 applications of an absolute pressure variation equivalent to approximately 20,000 feet pressure altitude. After 500, 1000 and 1500 cycles the altimeter shall be subjected to 2 cycles of temperature shock procedure I of MIL-E-5272 except that the high temperature shall be +71°C. in lieu of +85°C. Each pressure cycle shall consist of decreasing the pressure at a rate of 6,000 ± 2000 fpm until 20,000 ± 2000 feet is reached. The pressure shall be maintained at this pressure altitude for no less than 2 minutes. The pressure shall then be increased at a rate of 6000 ± 2000 fpm until ambient pressure altitude is reached. After a minimum of 2 minutes at ambient pressure the next cycle shall be commenced. This is to be continued until 2000 cycles have been completed after which the altimeter shall be rested at ambient pressure for approximately 48 hours.

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The contractor may remove the altimeter at any desired time during the cycling process to determine the altimeter performance at that time. Following completion of all cycling and the 48-hour rest period the altimeter may be reset by means of the zero setting adjustment mechanism. The amount of reset shall be limited to 200 feet. The instruments shall then be again subjected to and meet the requirements for all individual performance tests. In addition, the room temperature scale errors shall not deviate from the original scale error readings by more than the values specified in table V.

4.3.1.5.1 Following completion of the above tests the altimeter shall be placed in storage for a period of not less than 30 days. During the 30-day period the instrument shall remain as dormant as practical. The mechanism shall not be exercised other than that caused by normal ambient barometric pressure variations. The temperature shall remain essentially constant, however, normal room temperature variations are permissable. After the 30-day period, the altimeter shall again be subjected to and shall meet the requirements of the individual performance tests. In addition to meeting the tolerances specified in table II for scale error, the readings shall not vary from the original scale error readings by more than the values specified in table V.

TABLE V.- Scale error variation
(from readings taken in test specified in 4.3.1.8)

Standard altitude	Tolerance (± feet)
2000	125
5000	125
10,000	125
20,000	200

4.3.1.6 Zero setting system adjustment.- The altimeter shall be secured to prevent rotation and a suitable torque registering device shall be affixed to the center mounted setting knob. The knob shall then be rotated by application of force through the device in first one direction and then the other. The measured torque shall be recorded in inch-ounces.

4.3.1.7 Static balance error.- An altimeter reading shall be taken with the altimeter held in any desired position. The amount of vibration or tapping shall be sufficient to remove scale error due to friction effects. This reading shall not differ by more than 100 feet from the reading taken when the altimeter is held in any other position.

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4.3.1.8 Scale error at room temperature.- Testing shall be as follows: Set the pointer at zero using the zero setting system. Without changing the zero setting, subject the altimeter to successive changes in pressure corresponding to the pressure differentials specified in table VI. Compare the readings with a standard mercury barometer, and record any scale errors. The rate of change in pressure shall correspond to a rate of change in altitude of approximately 3,000 feet per minute. The altimeter shall remain at each pressure test point for not less than one minute and not more than ten minutes before a reading is taken.

TABLE VI.- Pressure differentials

Standard altitude (feet)	Equivalent atmospheric pressure (mercury)		Pressure differential (mercury)	
	mm	in	mm	in
0	760.00	29.92	. . .	. . .
2,000	706.65	27.82	53.35	2.10
5,000	632.36	24.90	127.64	5.02
10,000	522.65	20.58	237.35	9.34
20,000	349.25	13.75	410.75	16.17

4.3.1.9 Pointer movement.- The pointer movement shall be observed for any evidence of jump or backlash during all of the tests where pressure is varied at a uniform rate.

4.3.1.10 Friction error.- Testing shall be at each of the test points listed in table III. The pressure shall be decreased to bring the pointer approximately to the desired reading, then held constant while two readings are taken; the first before the altimeter is vibrated or tapped; and the second after vibrating or tapping. Any difference between the two readings, at each test point, shall be recorded as friction scale error, and shall not exceed the tolerances listed in table III. This test may be combined with the test specified in 4.3.1.8.

4.3.1.11 Hysteresis error.- Not more than 15 minutes after the altimeter has been subjected to the pressure corresponding to the upper limit of the scale in test specified in 4.3.1.8, the pressure shall be increased at a rate corresponding to a rate of decrease in altitude of 3,000 feet per minute until a pressure corresponding to 10,000 feet is reached. A reading shall be taken at this point and compared with the reading taken for the same point (ascending) in the test specified in 4.3.1.8. The above procedure shall be repeated at a pressure corresponding to 5,000 feet and 2,000 feet. The altimeter shall remain at each pressure test point for not less than a minute and not more than ten minutes before a reading is taken.

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4.3.1.12 Drift error.- Not less than one minute or more than five minutes after the completion of the hysteresis test, any difference between pointer indication and the zero setting specified in 4.3.1.8 shall be recorded. Correction, if required, shall be made for any change in atmospheric pressure.

4.3.1.13 Scale error at -55°C.- The test specified in 4.3.1.8 shall be repeated, with the difference that the temperature of the altimeter shall be stabilized at -55° ± 2°C. and the test performed at that temperature.

4.3.1.14 Scale error at +71°C.- The test specified in 4.3.1.8 shall be repeated, with the difference that the temperature of the altimeter shall be stabilized at +71°C. ± 2°C. and the test performed at that temperature.

4.3.1.15 Damping.- The reading of the altimeter shall be noted at atmospheric pressure. A vacuum sufficient to produce a pointer displacement of 360° shall be applied to the altimeter. The vacuum shall be instantaneously released and the pointer allowed to return to its original position. The return of the pointer shall be timed with a stop watch. The time, from release of vacuum until pointer passes the 3,000 foot indication, and from release of vacuum until pointer oscillation ceases shall be recorded.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (6.2).

5.1.1 Level A.- Each altimeter shall be preserved by method IId or IIf of MIL-P-116. The initial wrap shall consist of a close-fitting polyethylene bag. The bag opening shall be located at the smallest dimension. The film shall be minimum 0.001-inch thick, conforming to type II, grade A, B, or C, finish 1 of L-P-378. All seams shall be heat-sealed. Bag closure shall be effected by heat-sealing or double-folding the plastic material at the mouth and taping it to the bag body with pressure sensitive tape. All seams and closures effected by heat-sealing shall conform to the heat-seal strength requirements of L-P-378. The bagged altimeter shall be protected on all surfaces with a minimum of one-inch of cushioning material conforming to type II, Class B of PPP-C-843. Metal cans for method IId shall conform to type III of PPP-C-96, with suitable corrosion resistant key, and coated in accordance with plan B. Fiber cans for method IIf shall conform to type I, grade B, style b, class 1 of MIL-C-3955 with metal ends painted olive drab color.

5.1.2 Level C.- Altimeters shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.- Packing shall be level A, B, or C as specified (6.2).

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5.2.1 Level A.- Twelve altimeters, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V3c of PPP-B-636. Level A packages shall be packed upright three in length, two in width, and two in depth within the container. Each shipping container shall be provided with a top and bottom pad and a layer separator pad made of the same material as the shipping container. The length and width dimensions of each pad shall be one-eighth of an inch less than the inside length and width of the container. Each shipping container shall be closed and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2 Level B.- Unless otherwise specified (see 5.2.2.1), twelve altimeters, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety optional, class domestic, grade 200 of PPP-B-636. Level A packages shall be packed three in length, two in width, and two in depth within the container. Each shipping container shall be provided with pads as specified in 5.2.1, except that the fiberboard shall be class domestic, grade 200. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2) fiberboard shipping containers shall be made from weather-resistant fiberboard as specified in 5.2.1.

5.2.3 Level C.- Altimeters, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking.- The following precautionary marking shall appear on two opposite sides of each interior package and shipping container:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use.- The altimeter covered by this specification is intended for use by parachutists in determining altitude in free fall, delayed opening parachute jumps. The altimeter is a component of the High Altitude Low Opening (HALO) parachute system.

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6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Whether preproduction sample is required, and if required, number of samples (see 3.1).
c. Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).
d. Whether weather-resistant class fiberboard shipping containers are required for level B shipping (see 5.2.2.1).

6.3 Acceptable cover-glass materials.- The following materials have been found acceptable for the cover-glass application:

a. "Lexan" - General Electric Co.
b. "Merlon" - Mobay Chemical Co.

Equivalent material may be used subject to the approval of the procuring activity.

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Army - GL

Review activities:

Army - AV, MD

Preparing activity:

Army - GL

Project No. 6610-A005
Code "C"

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Documents.

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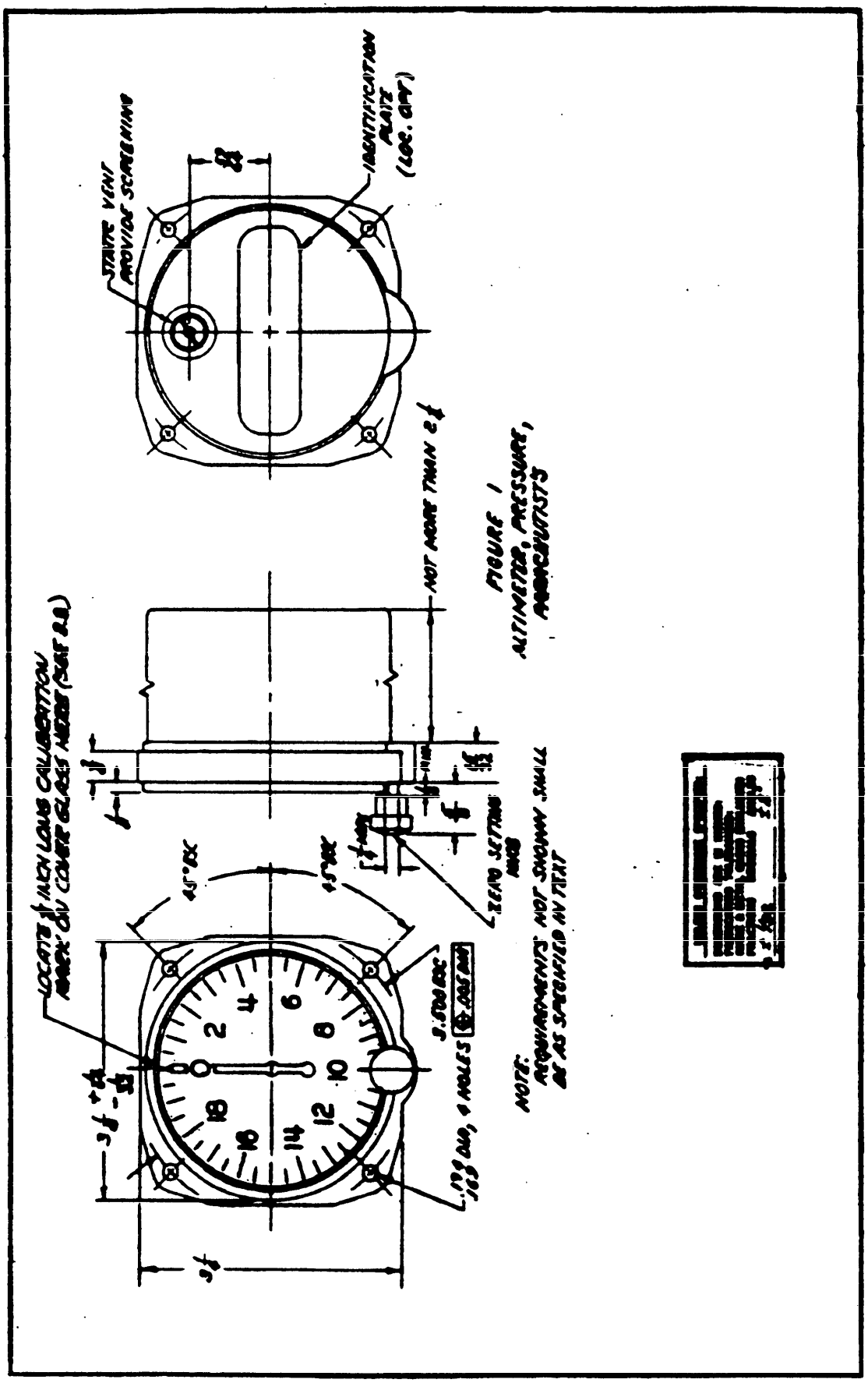


FIGURE 1
ALTIMETER, PRESSURE, AERONAUTICAL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Altimeter, Pressure, Parachutist's MIL-S-58066A(GL) - FSC 6610

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

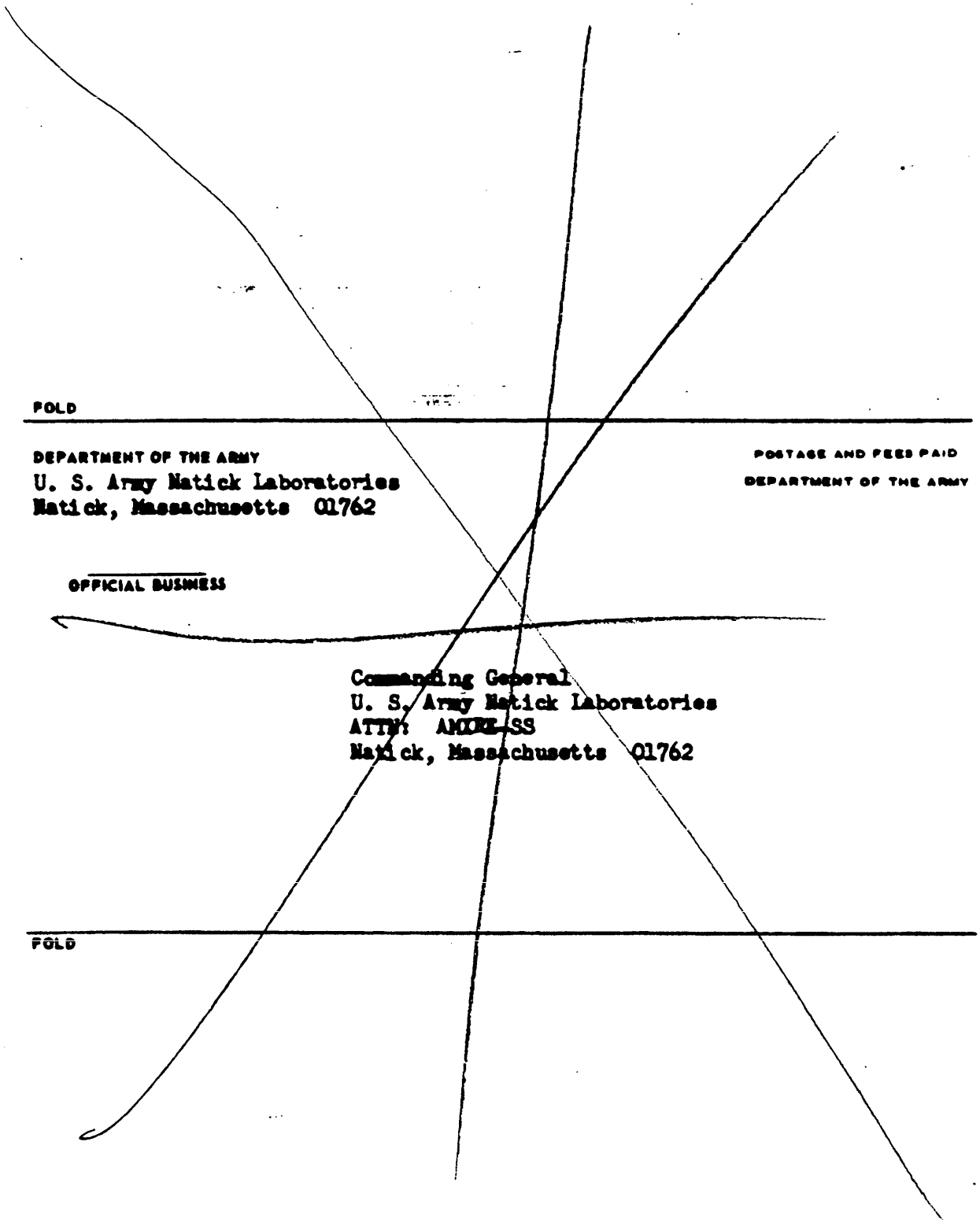
☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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